The Art of Queueing

Reinforcement Learning for visitor flow at the Gallerie degli Uffizi in Florence

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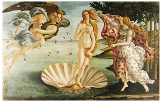
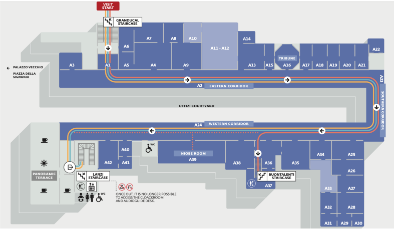
The few rooms everyone queues for

Almost all ∼5,000 daily visitors come for the same handful of paintings, crammed into three of ninety-eight rooms, so those rooms run far over capacity while the rest sit calm. The crowd is mixed: *60%* Instagram tourists, *30%* normal, *10%* art lovers.

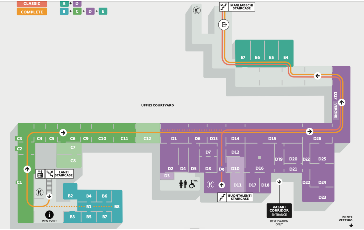





Second floor (the gated masterpieces)



First floor (Caravaggio, free-flow)



Two streams, one project

We split the project into two parts kept separate until the end.

Stream 1: crowd simulator.

Full 60/30/10 crowd, one day.

- Museum *as-is* versus a Pigovian redesign.
- Adopt only if welfare rises, revenue holds.
- It passes: welfare $+31\%$, revenue preserved.

Stream 2: RL on top (our focus).

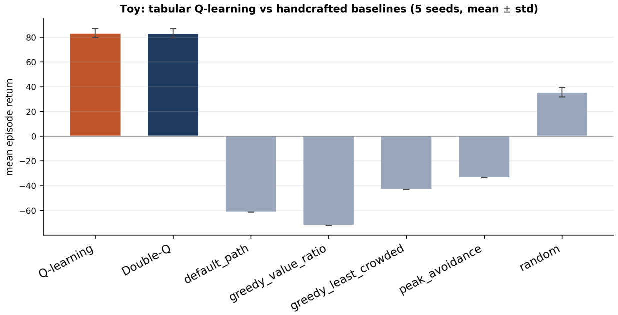
Take that museum as fixed; find the *optimal* visitor.

- When to book and which masterpieces.
- How to pace the day.
- All under crowd uncertainty.

Gate first, then the optimal visit.

Foundations: a tabular proof of concept

We prove it on a *12-room toy* with an exact value table. Crowds wave up and down, unannounced.
Tabular *Q-learning* learns *temporal arbitrage*: enter in the lull around the crowd peak.
Across five seeds: Q and *Double-Q tie* (83.3 vs 83.1). A deterministic toy has no overestimation bias to remove, exactly as theory predicts.



Learning beats every hand-written rule; the rules even go negative by mistiming the crowd.

Timing, learned from samples, beats every hand-written rule.

Scaling up: deep RL and action masking

The real museum breaks the table: ninety-eight rooms, $\sim$100 moves per step, hundreds of state numbers. We swap it for a neural net (*MaskablePPO*; PPO and DQN as controls).

Action masking: the rulebook

From any room only a few moves are legal, yet the net has $\sim$100 outputs. We set illegal moves to $-\infty$ before the softmax: zero probability, no gradient wasted on them. This state-dependent action set is our single biggest design choice.

Decisive exactly where the decision is hardest.

The reframe: when to book, not where to walk

A visitor has a map and guide, so the route is a shortest path. The real decision is the *booking*, hard for three reasons walking never has:

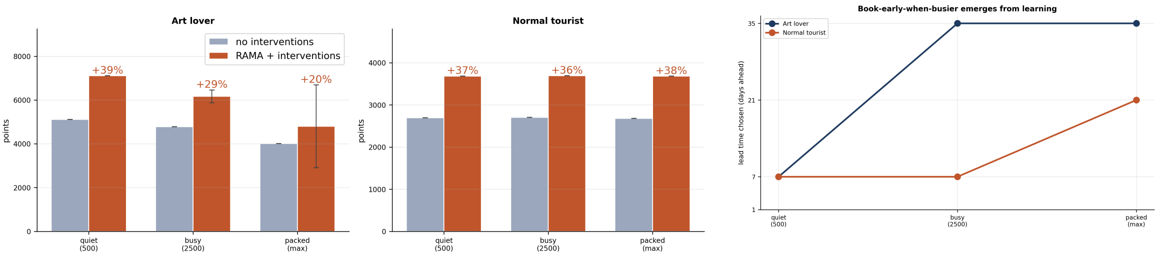
- *Uncertain:* you cannot see how fast the good slots fill.
- *Irreversible:* book late and the masterpiece is gone.
- *Delayed:* you act today; the payoff lands at check-in ($\gamma = 0.999$).

Baseline (as-is): no booking, take masterpieces as found.

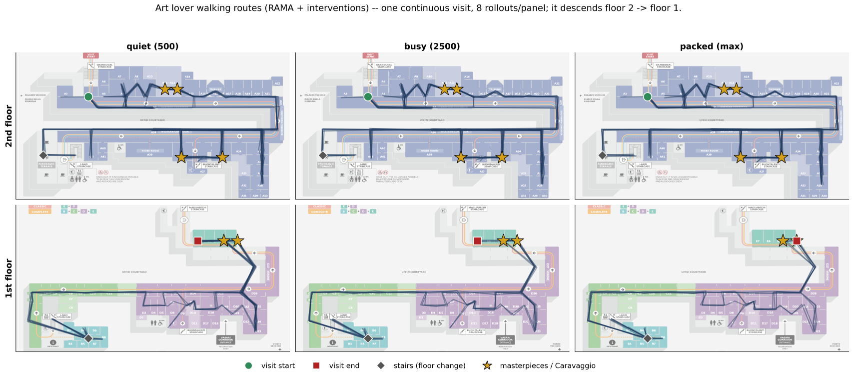
RAMA: booked slots, pick a lead time of 1, 7, 21 or 35 days.

Result: book early when busier and it pays

RAMA booking: intervened vs matched baseline (5 seeds, mean ± std; 3/3 masterpieces secured)



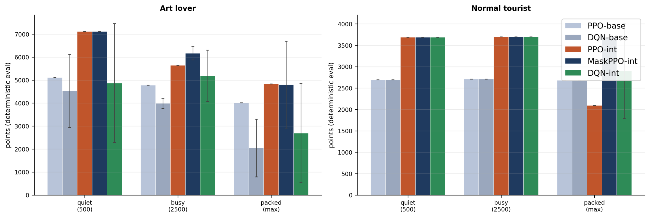
Both profiles beat the matched no-booking baseline at every crowd level (five seeds, mean$\pm$std): +20% to +39% for the art lover, +36% to +38% for the tourist, with all three masterpieces secured. The learned lead time *rises with the crowd* (right): book-early-when-busier is not coded in, it emerges from the reward, because booking late at a packed museum misses the rooms.



The art lover's learned visit, eight rollouts per panel, across the three crowd levels. It is one continuous, sensible route: enter at `A1`, sweep the second-floor masterpieces (gold stars) in their booked windows, descend the staircase to Caravaggio on the first floor and leave before closing. Every step is a one-hop move to a neighbouring room, so there are no teleports.

Which algorithm wins and why masking matters

Algorithm comparison: value-based (DQN) vs policy-gradient (PPO / MaskablePPO), baseline vs intervened (5 seeds, mean ± std)



- *MaskablePPO* is best or tied in every cell and stable across seeds.
- *Masking is decisive exactly at the hardest cell:* at the packed tourist, unmasked PPO *collapses* to 2093, below the no-booking baseline of 2680, while MaskablePPO holds 3685. At easier crowds the two agree; the gap appears only where precision matters.
- *DQN is crowd-fragile and high-variance* (its baseline falls 4527 $\rightarrow$ 2044): taking a max over noisy crowd values overestimates, the very bias the deterministic toy did not have.

1: an unlearnable reward

S *Symptom.* The agent never left, squatting in one room until closing. Zeroing all reward for not exiting made it *worse*.

D *Diagnosis.* An all-or-nothing wipe is *flat* almost everywhere, so the gradient is zero: nothing points to the door.

F *Fix.* Use a *dense* egress signal: per-step pressure rising near closing and far from an exit, plus a bonus for leaving.

L *Lesson.* Sparse rewards are correct but often unlearnable. Shape a slope the optimiser can climb.

2: it could not see what it was deciding

S *Symptom.* The rule “stay until satisfied, then move” never stabilised; dwell times wobbled at random.

D *Diagnosis.* The right action depended on value already taken from the room, *not in the state*: a hidden variable, a POMDP not the MDP we assumed.

F *Fix.* Add a per-room *appreciation-progress* value to the state, so “seen enough?” becomes something the policy can read.

L *Lesson.* The state must hold everything the optimal action depends on, or you solve a different problem.

3: reward shaping that backfired

S *Symptom.* Well-meant terms backfired: a “boredom” penalty caused *looping*, a harsh crowd penalty made a *fake art lover*.

D *Diagnosis.* Every term is a new objective to game (reward hacking): looping beat exiting, a crowded Botticelli scored worse than skipping it.

F *Fix. Minimal* shaping: a *gentle* crowd discount so a crowded masterpiece still beats skipping, no boredom penalty, satiation alone stops lingering.

L *Lesson.* Ask of every term: what is the cheapest way to collect it? Is that what you want?

4: the do-nothing trap and the book-early trick

S *Symptom.* Given a free "decline" button, *both* profiles declined all three masterpieces; book-early stayed only *half* learned.

D *Diagnosis.* Declining is safe now, booking pays only later: a do-nothing optimum; its reward lands hundreds of steps away (credit assignment).

F *Fix.* Make "always books" a *type trait* with no escape hatch; add an *immediate off-pace penalty* proxying the distant cost.

L *Lesson.* A small immediate signal tracking a far-off reward makes the long-horizon lesson learnable: our most decisive trick.

What five seeds revealed (and why we ran them)

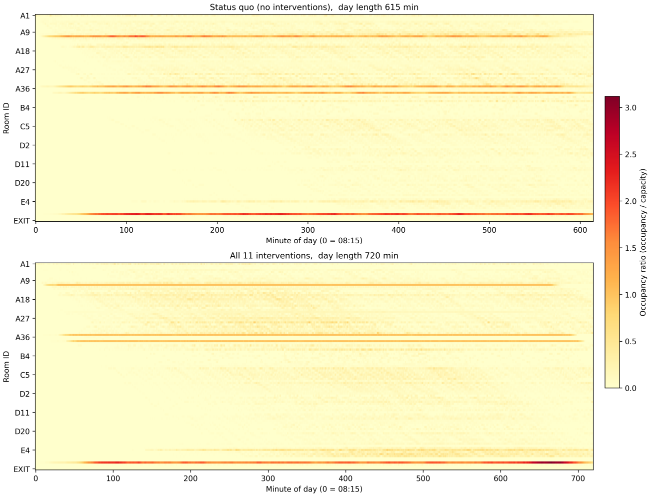
A single training run can be lucky or unlucky, so a single number can mislead. We retrained every cell under *five* training seeds and report the mean and the spread across them.

- *The tourist is rock-stable*: essentially zero variance at every crowd level.
- *The art lover degrades as the museum fills*: the gain falls from +39% to +29% to +20%.
- *The catch:* at *maximum crowd* the art lover is *bimodal*, four of the five seeds reach ∼+41% but one collapses below baseline, so the cell averages +20% with a wide error bar. We report the spread, not the lucky seed.

Reporting the spread is part of the result, not a footnote.

The crowd before and after the redesign

Crowd density heatmap: status quo vs all 11 interventions (~5000 visitors). Shared colour scale.



Top: the museum as-is. *Bottom:* all eleven interventions, same colour scale.

The masterpiece bands (`A9`, `A36`) cool from red toward orange and the day spreads over more hours.

+31% welfare, revenue held.

What worked, what did not, what is debatable

Worked best

- *Action masking*, the biggest lever.
- *Reframing* navigation into booking.
- *An immediate proxy* for a delayed reward.
- *Exposing appreciation progress*.
- *Dense egress shaping*: a way out.

Worked worst

- *All-or-nothing exit wipe*: zero gradient.
- *A free decline button*: a do-nothing trap.
- *A harsh crowd penalty*: a fake art lover.
- *A boredom penalty*: perverse looping.
- *Greedy evaluation* of a stochastic policy.

Debatable: the 60/30/10 split and the dwell and crowd calibration are modelling choices; the fill curve is calibrated, not fitted; the as-is museum hides the real entry choice (book or queue).

Thank you.

Questions?